Parameter Estimation: 20 Worked Problems
MLE · Method of Moments · Traps · German Tank/MVUE · Bias/Var/MSE · MLE Properties

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Classical estimation is where probability theory turns into a decision: given data, what's your single best guess for an unknown parameter, and how do you even judge "best"? This sheet builds the full toolkit — MLE, Method of Moments, the German Tank Problem's famous MVUE correction, and the Bias/Variance/MSE framework that lets you compare estimators rigorously — with color-coded sections and an analogy on every problem. Several problems reuse the *same* dataset across techniques deliberately, so you can watch MLE, Method of Moments, and the MVUE disagree (and see exactly why) on identical data.

I. Maximum Likelihood & Method of Moments

Problem A1 — MLE for a Binomial proportion A market-making algorithm sends $n=20$ quotes; $k=14$ get filled. Find the MLE of the fill probability p.

Solution.

$$L(p)=\binom{20}{14}p^{14}(1-p)^{6}.$$

Maximizing the log-likelihood: $\frac{d}{dp}\left[14\ln p+6\ln(1-p)\right)=\frac{14}{p}-\frac{6}{1-p}=0\Longrightarrow 14(1-p)=6p$

$$\Longrightarrow \hat{p}=\frac{14}{20}=0.7\,.$$

★ *Intuition:* The MLE for a Binomial proportion is always just "successes over trials" — the calculus always collapses to the obvious sample proportion. The real value of grinding through the derivative once is trusting the shortcut forever after.

Problem A2 — MLE for a Poisson rate A desk logs the number of large block trades on 10 independent days: $3,5,2,4,6,3,5,4,2,6$. Find the MLE of λ.

Solution. For Poisson, the MLE is always the sample mean:

$$\hat{\lambda}=\acute{x}=\frac{3+5+2+4+6+3+5+4+2+6}{10}=\frac{40}{10}=4.$$

★ *Intuition:* Just like A1, the scary-looking log-likelihood maximization for Poisson always resolves to "just average your data" — recognizing *which* distributions reduce to trivial sample statistics (Binomial → proportion, Poisson → mean) is worth more at Stage-1 speed than re-deriving any of them from scratch.

Problem A3 — MLE for discrete Uniform (one observation): the Locomotive setup A railroad numbers its locomotives $1, \ldots, N$. You see one locomotive, numbered 60. Find the MLE of N.

Solution. $L(N) = \frac{1}{N}$ for every $N \geq 60$, and 0 otherwise — a *decreasing* function of N on its support. So $L(N)$ is maximized by taking N as *small* as the data allow:

$$\widehat{N}_{\text{MLE}} = 60.$$

★ *Intuition:* This is deliberately weird: the likelihood doesn't peak at an interior point like A1/A2 — it's maximized right at the boundary of what's logically possible, since any $N < 60$ is instantly ruled out (impossible to observe locomotive 60 if only 59 exist) while every $N \geq 60$ remains merely *less and less likely* as N grows. This boundary behavior is the entire reason the German Tank Problem (Section II) needs a special correction that ordinary MLE doesn't.

Problem A4 — MLE for a Geometric success probability Five independent large orders took $4, 7, 3, 6, 5$ order-book snapshots respectively before filling. Find the MLE of p.

Solution. For Geometric(p) on $\{1, 2, \ldots\}$, $L(p) = p^n (1-p)^{\sum x_i - n}$. Maximizing gives $\hat{p} = n / \sum x_i = 1/\acute{x}$:

$$\acute{x} = \frac{4+7+3+6+5}{5} = 5 \Longrightarrow \hat{p} = \frac{1}{5} = 0.2.$$

★ *Intuition:* "MLE ¿ $1/\acute{x}$" makes immediate sense: if it typically takes 5 attempts to succeed, your best estimate of the per-attempt success rate is $1/5$ — the MLE machinery here just formalizes an estimate you'd write down by instinct.

Problem A5 — MLE for discrete Uniform (multiple observations): setting up the Tank Problem 5 captured tanks have serial numbers $19, 42, 7, 33, 51$ (drawn uniformly without replacement from $\{1, \ldots, N\}$). Find the MLE of N.

Solution. Exactly as in A3, $L(N) = \binom{N}{5}^{-1}$ for $N \geq max(\text{data})$, decreasing in N — maximized at the smallest admissible N:

$$\widehat{N}_{\text{MLE}} = max(19, 42, 7, 33, 51) = 51.$$

★ *Intuition:* Notice the MLE threw away 4 of the 5 data points entirely — only the maximum mattered. That feels wasteful, and it is: Section II exists precisely because this throws away useful information and produces a systematically-too-low estimate.

Problem B1 — Method of Moments for Poisson (and why it agrees with MLE) Using A2's data (sample mean $\acute{x} = 4$), find the Method of Moments estimator of λ.

Solution. MoM sets the population mean equal to the sample mean. Since $E[\text{Pois}(\lambda)] = \lambda$:

$$\hat{\lambda}_{\text{MoM}} = \acute{x} = 4 \left(\text{identical to the MLE from A2}\right).$$

★ *Intuition:* For many "nice" one-parameter families (Poisson, Binomial with known n), MoM and MLE coincide exactly, because the single population moment *is* the natural parameter. The two methods only start to diverge — interestingly — once the parameter isn't directly the mean, as the very next problem shows.

Problem B2 — Method of Moments for discrete Uniform: a genuine disagreement with MLE Using A5's tank data (serials $19, 42, 7, 33, 51$; sample mean ¿ 30.4), find the Method of Moments estimator of N.

Solution. For discrete Uniform$\{1, \ldots, N\}$, $E[X] = \frac{N+1}{2}$. Set this equal to the sample mean:

$$\frac{N+1}{2} = 30.4 \Longrightarrow \widehat{N}_{\text{MoM}} = 2\left(30.4\right) - 1 = 59.8.$$

★ *Intuition:* Now compare to A5's MLE of 51: MoM uses *every* data point (via the mean) rather than just the maximum, and produces a noticeably larger, more plausible-feeling estimate. This is the clearest illustration in the whole sheet of MLE and MoM genuinely disagreeing — and it's not obvious yet which one (if either!) is actually better. That's exactly what Sections II–III resolve.

Problem B3 — Two-parameter Method of Moments Monthly trading halts are modeled as Binomial$\left(n, p\right)$ with *both* n and p unknown. Historical data gives sample mean $\acute{x} = 6$ and sample variance $s^2 = 4.2$. Find the MoM estimates of n and p.

Solution. Match both moments: $n\,p = 6$ and $n\,p\left(1 - p\right) = 4.2$. Dividing the second by the first:

$$1 - p = \frac{4.2}{6} = 0.7 \Longrightarrow \hat{p} = 0.3, \hat{n} = \frac{6}{0.3} = 20.$$

★ *Intuition:* One-parameter MoM only needs the mean; the instant you have *two* unknowns, you need *two* moments (mean *and* variance) to pin them both down — match as many moments as you have unknowns, no more, no fewer. This generalizes directly to MoM for any k-parameter family.

Problem C1 — MLE vs. MoM, side by side: which one do you trust? Using the tank data throughout (max ¿ 51, mean ¿ 30.4, $n = 5$), you now have MLE ¿ 51 and MoM ¿ 59.8. Which is more reasonable, and why might *both* still be flawed?

Solution. MLE (51) systematically **ignores all information below the maximum** and is provably biased low (Section II). MoM (59.8) uses the full sample, but has no mechanism to *guarantee* its estimate even respects the constraint $N \geq max\left(\text{data}\right) = 51$ — it got lucky here, but won't always (see C2). Neither is designed to be the *minimum-variance unbiased* estimator; that requires the dedicated correction in Section II.

★ *Intuition:* Think of MLE as a stubborn minimalist (trusts only the single most extreme data point) and MoM as an average-truster (uses everything, but blindly, with no boundary

awareness). The MVUE, coming next, is the estimator that's actually disciplined about both using all the data *and* respecting the problem's hard constraints.

Problem C2 — The trap: MoM can give a logically impossible estimate Capture-recapture data: 5 tagged numbers are $2, 3, 1, 50, 4$ (one outlier). Compute MLE and MoM for N, and explain what's gone wrong.

Solution. $max = 50$, mean ¿ 12.

$$\widehat{N}_{\text{MLE}} = 50, \widehat{N}_{\text{MoM}} = 2(12) - 1 = 23.$$

But $23 < 50 = max(\text{data})$ — the MoM estimate claims fewer tanks exist than we've already directly observed! This is **logically impossible**, since N must be at least as large as the largest serial number ever seen.

★ *Intuition:* MLE, whatever its other flaws, can *never* produce an impossible estimate — it's constructed to always satisfy $\widehat{N} \geq max(\text{data})$ by definition. MoM has no such built-in safety rail and can be derailed by a single outlier. This is a genuinely important lesson for real-world estimation: an estimator's *elegant derivation* doesn't guarantee it respects the problem's hard constraints — always sanity-check MoM output against logical bounds before trusting it.

▸ *Speed trick:* If a MoM estimate ever violates a hard structural constraint (like $\widehat{N} < max(\text{data})$ here), that's an instant red flag to fall back to MLE or a dedicated unbiased estimator — never report a logically impossible number just because a formula produced it.

II. The German Tank Problem & the MVUE

Problem D1 — Quantifying the MLE's bias For $N = 100$ true tanks, $k = 5$ captured, find $E[m]$ (the expected sample maximum) and hence the bias of the naive MLE $\widehat{N} = m$.

Solution.

$$E[m] = \frac{k(N+1)}{k+1} = \frac{5(101)}{6} = \frac{505}{6} \approx 84.17.$$

$$\text{Bias} = E[m] - N = 84.17 - 100 = -15.83.$$

★ *Intuition:* With 100 true tanks and only 5 captured, the MLE systematically reports "about 84" — underestimating by nearly 16 tanks *on average*, not just occasionally. This isn't bad luck in one sample; it's a structural, provable bias built into using the maximum alone.

Problem D2 — Deriving and verifying the MVUE correction Show that $\widehat{N} = m + \frac{m}{k} - 1$ is unbiased, then apply it to the tank data (max ¿ 51, $k = 5$).

Solution. Unbiasedness. Using $E[m]=\frac{k(N+1)}{k+1}$:

$$E[\widehat{N}]=E[m]\left(1+\frac{1}{k}\right)-1=\frac{k(N+1)}{k+1}\cdot\frac{k+1}{k}-1=(N+1)-1=N.\checkmark$$

Application:

$$\widehat{N}=51+\frac{51}{5}-1=51+10.2-1=60.2.$$

⋆ *Intuition:* Compare all three estimates on identical data now: MLE ¿51 (too low), MoM ¿59.8 (lucky, close), MVUE ¿60.2 (provably unbiased, and notice how close it lands to MoM here — reassuring, but MVUE is the one with a guarantee, not a coincidence).

Problem D3 — Reproducing Mosteller's own answers exactly Apply the MVUE formula to Mosteller's classic Locomotive Problem: (a) one locomotive observed, numbered 60 ($k=1$); (b) five locomotives observed, max ¿60 ($k=5$).

Solution.

$$\text{(a) }\widehat{N}=60+\frac{60}{1}-1=60+60-1=119.$$

$$\text{(b) }\widehat{N}=60+\frac{60}{5}-1=60+12-1=71.$$

⋆ *Intuition:* These exactly match Mosteller's own published answers (119 and 71) to this very problem — the MVUE formula isn't an abstract exercise, it's the literal machinery behind one of the most famous worked examples in elementary statistics, applied directly to the source material your syllabus is built on.

Problem D4 — The MVUE's variance Derive $\mathrm{Var}(\widehat{N})$ from $\mathrm{Var}(m)=\frac{k(N-k)(N+1)}{(k+1)^2(k+2)}$, and evaluate at $N=100$, $k=5$.

Solution. Since $\widehat{N}=\frac{k+1}{k}m-1$ is a linear function of m:

$$\mathrm{Var}(\widehat{N})=\left(\frac{k+1}{k}\right)^2\mathrm{Var}(m)=\frac{(N-k)(N+1)}{k(k+2)}.$$

At $N=100$, $k=5$: $\mathrm{Var}(\widehat{N})=\frac{95(101)}{5(7)}=\frac{9595}{35}\approx 274.1$ (sd ≈ 16.6).

⋆ *Intuition:* Even though $\widehat{N}$ is perfectly unbiased, it's not *precise* — with only 5 captured tanks out of 100, a standard deviation of ± 17 tanks is a genuinely wide net. Unbiasedness is a statement about the *average* behavior across many hypothetical samples, never a promise about how close any *one* estimate will land.

III. Bias, Variance & Mean Squared Error

Problem E1 — MSE of the naive (biased) MLE Using D1's results ($N=100$, $k=5$: Bias ¿ -15.83, $\mathrm{Var}(m)\approx 190.38$), compute $\mathrm{MSE}\left(\widehat{N}_{\mathrm{MLE}}\right)$.

Solution.

$$\mathrm{MSE}=\mathrm{Bias}^2+\mathrm{Var}=(-15.83)^2+190.38=250.7+190.38\approx 441.1.$$

★ *Intuition:* MSE punishes bias *quadratically* — a modest-looking bias of -15.83 contributes over 250 to the MSE all by itself, more than the variance term. This is why "just use the most likely single value" (MLE) can be a genuinely costly habit once you measure total error correctly.

Problem E2 — MSE showdown: MLE vs. MVUE Compare $\mathrm{MSE}\left(\widehat{N}_{\mathrm{MLE}}\right)\approx 441.1$ (E1) to $\mathrm{MSE}\left(\widehat{N}_{\mathrm{MVUE}}\right)$ at the same $N=100$, $k=5$.

Solution. The MVUE is unbiased, so its MSE is *purely* its variance (D4):

$$\mathrm{MSE}\left(\widehat{N}_{\mathrm{MVUE}}\right)=\mathrm{Var}\left(\widehat{N}\right)\approx 274.1.$$

$$441.1\,(\mathrm{MLE})\ \text{vs.}\ 274.1\,(\mathrm{MVUE})\Longrightarrow \text{MVUE wins by}\approx 38\%.$$

★ *Intuition:* Here the MVUE doesn't just win on principle — it wins decisively on the one number (MSE) that actually matters for practical accuracy. This is the payoff for the extra derivation work in Section II: a 38% reduction in total expected squared error, for free, just by using all the data correctly instead of only the maximum.

Problem E3 — The real bias-variance trade-off: a biased estimator can win For $\mathrm{Binomial}(n=10, p)$ with true $p=0.3$, compare the (unbiased) MLE $\hat{p}=k/n$ to the biased "shrinkage" estimator $\tilde{p}=\dfrac{k+1}{n+2}$.

Solution. MLE: unbiased, $\mathrm{Var}=\dfrac{p(1-p)}{n}=\dfrac{0.21}{10}=0.021=\mathrm{MSE}$. **Shrinkage:**

$E[\tilde{p}]=\dfrac{np+1}{n+2}=\dfrac{4}{12}=1/3$, so $\mathrm{Bias}=1/3-0.3=1/30\approx 0.0333$.

$$\mathrm{Var}(\tilde{p})=\frac{np(1-p)}{(n+2)^2}=\frac{2.1}{144}\approx 0.01458.$$

$$\mathrm{MSE}(\tilde{p})=(0.0333)^2+0.01458\approx 0.00111+0.01458=0.01569.$$

$$0.01569\,(\text{shrinkage})<0.021\,(\mathrm{MLE})!$$

★ *Intuition:* This is the genuine bias-variance trade-off, not just "unbiased always wins" (as Section II might have suggested). The shrinkage estimator deliberately pulls toward 0.5, accepting a small bias in exchange for a larger reduction in variance (dividing by $(n+2)^2$

instead of n) — and the trade pays off here. This exact mechanism (shrinkage toward a central value, trading bias for variance) is the foundational idea behind ridge regression, Bayesian priors, and most of modern regularized estimation.

Problem E4 — How the MLE/MVUE gap evolves with sample size Repeat the $N=100$ MSE comparison (E2) at $k=20$ instead of $k=5$, and describe the trend.

Solution.

$$\text{At } k=20: \text{MSE(MLE)} = \frac{2400}{77} \approx 31.2, \text{MSE(MVUE)} = \frac{202}{11} \approx 18.4.$$

$$\text{Ratio MLE/MVUE:} \approx 1.61 \,(k=5) \text{ vs.} \approx 1.70 \,(k=20).$$

★ *Intuition:* Both estimators improve enormously as k grows (MSE drops by more than 10 × from $k=5$ to $k=20$) — more data helps everyone. But the *relative* gap between MLE and MVUE doesn't close; if anything it's slightly wider at $k=20$. The lesson: switching to the better estimator (MVUE) and collecting more data are two *independent* levers, and neither substitutes for the other.

IV. Bonus: MLE Properties (your-side extension)

Problem F1 — The invariance property Given the MLE $\hat{p}=0.7$ from A1, find the MLE of the *odds* $\theta = \frac{p}{1-p}$ — without re-deriving anything from scratch.

Solution. Invariance property: if $\hat{\theta}$ is the MLE of a parameter, then $g(\hat{\theta})$ is automatically the MLE of $g(\theta)$, for *any* function g. Here $g(p) = \frac{p}{1-p}$:

$$\hat{\theta}_{\text{odds}} = \frac{\hat{p}}{1-\hat{p}} = \frac{0.7}{0.3} = 7/3 \approx 2.33.$$

★ *Intuition:* Without invariance, you'd need to re-derive an entirely new likelihood function every time you cared about a transformed quantity (odds, log-odds, variance instead of sd, …). Invariance means you only ever need to find the MLE of the *original* parameter once, then plug it into whatever function of interest you like — an enormous, under-appreciated time-saver.

Problem F2 — Sufficiency: why the count is all you need Two different 20-quote sequences both have $k=14$ fills. Show the likelihood ratio between any two candidate values p_1, p_2 depends on the data *only* through k — meaning no information is lost by recording just the fill count instead of the full fill/no-fill sequence.

Solution. For any specific sequence with k fills out of n attempts, $L(p) = p^k (1-p)^{n-k}$ (order doesn't enter the formula at all — every sequence with the same k has the *same* likelihood). So for two parameter values:

$$\frac{L(p_1)}{L(p_2)} = \frac{p_1^k(1-p_1)^{n-k}}{p_2^k(1-p_2)^{n-k}}$$

depends on the data *only* through k (and the fixed n) — never on *which* specific quotes filled.

★ *Intuition:* This is the formal notion of a **sufficient statistic**: k alone carries every ounce of information the full 20-length sequence carries about p. A trading desk's compliance log could discard the exact order of fills and keep only the daily fill *count*, and a statistician would lose absolutely nothing — which is exactly why every MLE derivation in this sheet (A1–A5) only ever needed a simple summary statistic (a sum, a mean, a max), never the raw sequence itself.

Drill order for review: re-derive D2 (MVUE unbiasedness), D3 (both Mosteller cases), and E3 (the shrinkage trade-off) from scratch, closed-book. Together they carry the chapter's central arc: MLE is simple but can be structurally biased (A3/A5), the bias is quantifiable and correctable (D1–D2), and MSE — not bias alone — is the only honest way to compare estimators (E1–E3), with E3 proving that "unbiased" and "best" are not synonyms.